Matthias Vancraeynest

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Education

Ph.D. in Mathematics — Edinburgh, Scotland
University of Edinburgh — 2023 - 2027 *(expected)*
- Advisors: Prof. David Jordan and Prof. Alexander Shapiro

Master of Mathematics: M2 de Mathématiques Fondamentales — Paris, France
Université Paris Cité — 2021-2022
- Thesis: *On the Kashiwara-Vergne Conjecture*, advised by Prof. Adrien Brochier

Master of Physics: Theoretical Physics — Leuven, Belgium
KU Leuven — 2019-2021
- Thesis: *Holographic backgrounds of low dimension*, advised by Prof. Thomas Van Riet
- Summa cum laude
- Exchange to the Université de Genève in the Spring semester 2020

Bachelor of Mathematics — Leuven, Belgium
KU Leuven — 2016-2019
- Minor Physics and Astronomy
- Magna cum laude

Publications

1. J. Brown, J. R. Gómez García, D. Jordan and M. Vancraeynest, *Weyl defects in skein theory and quantum cluster charts*, (preprint). ArXiv 2609.09308 (2026)
2. J. Bierent, D. Jordan, M. Vancraeynest and M. Vazirani, *The skein partition function of the mapping torus*, (preprint). ArXiv 2511.19139 (2025). An extended abstract appeared in the proceedings of FPSAC 2026.
3. R. C. Avohou, R. Toriumi and M. Vancraeynest, *Classification of higher grade ℓ graphs for $U(N)^2 \times O(D)$ multi-matrix models*, Annales de l'Institut Henri Poincaré D, Combinatorics, Physics and their Interactions **13** (2026), no. 2, pp. 281–355. DOI: 10.4171/aihpd/217.

Talks

Hodge Club, *University of Edinburgh*	Feb 2026
Quantum Universe Attract.Workshop, *University of Hamburg*	Nov 2025
GEARS seminar, *Heriot-Watt University*	Sep 2025
MIGS Student conference, *University of Edinburgh*	Mar 2025
GRIFT seminar, *University of Edinburgh*	Nov 2024
Hodge Club, *University of Edinburgh*	Oct 2024
SMSTC Hopf Algebras, *University of Glasgow*	Mar 2024
GlaMS example showcase, *University of Glasgow*	Dec 2023

Posters

Contact Geometry, Cluster Algebras and Skein Theory, *IPAM, Los Angeles*	Oct 2026
New perspectives on skein modules, *CIRM, Luminy*, with J. Bierent	Jul 2025

Experience

Educational developer — Brussels, Belgium
Vrije Universiteit Brussel — Feb - May 2023
- Student support service

Research intern — Okinawa, Japan
Okinawa Institute of Science and Technology — Oct 2022 - Jan 2023
- Gravity, Quantum Geometry and Field Theory Unit
- Advisor: Prof. Reiko Toriumi

Research visits

Institute for Pure and Applied Mathematics (IPAM), UCLA Los Angeles, USA
Quantum Topology, Character Varieties and Low-Dimensional Geometry Sep - Dec 2026

Teaching

University of Edinburgh

Introduction to Mathematical Analysis, Tutorials	Spring 2026
Mathsbase, Drop-in sessions	Spring 2026
Geometry, Tutorials	Autumn 2025
Mathsbase, Drop-in sessions	Autumn 2025
Introduction to Number Theory, Tutorials	Spring 2025
Fundamentals of Pure Mathematics, Tutorials	Spring 2025
Mathsbase, Drop-in sessions	Spring 2025
Introduction to Linear Algebra, Tutorials	Autumn 2024

Skills

Languages: Dutch, English, French.
Programming: `Python`, `MATLAB`, `R`.

Organisation

University of Edinburgh
Hodge Club, Postgraduate seminar 2024-2025